EXPERIMENT - 4

Implement packet sniffing using raw sockets in python

Aim - To implement a packet sniffer using Python Raw Sockets to capture and analyze network packets.

Requirements

Hardware

• Laptop/Desktop

• Minimum 4 GB RAM

Software

• Python 3.8 or above

• Windows 10/11 or Ubuntu Linux

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Packet_Sniffing.py - C:/Users/MAHATHI A J/Documents/PacketSniffer/Packet_Sniffing.py (3.14.0)

File Edit Format Run Options Window Help

```
import socket
import sys
try:
    s = socket.socket(
        socket.AF_INET,
        socket.SOCK_RAW,
        socket.IPPROTO_IP
    )

except PermissionError:
    print("Run this program as Administrator (Windows) or root (Linux).")
    sys.exit(1)

HOST = "127.0.0.1"
s.bind((HOST, 0))
s.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
try:
    s.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
except (AttributeError, OSError) as e:
    print("Promiscuous mode could not be enabled:", e)

print("Packet Sniffer Started...\n")

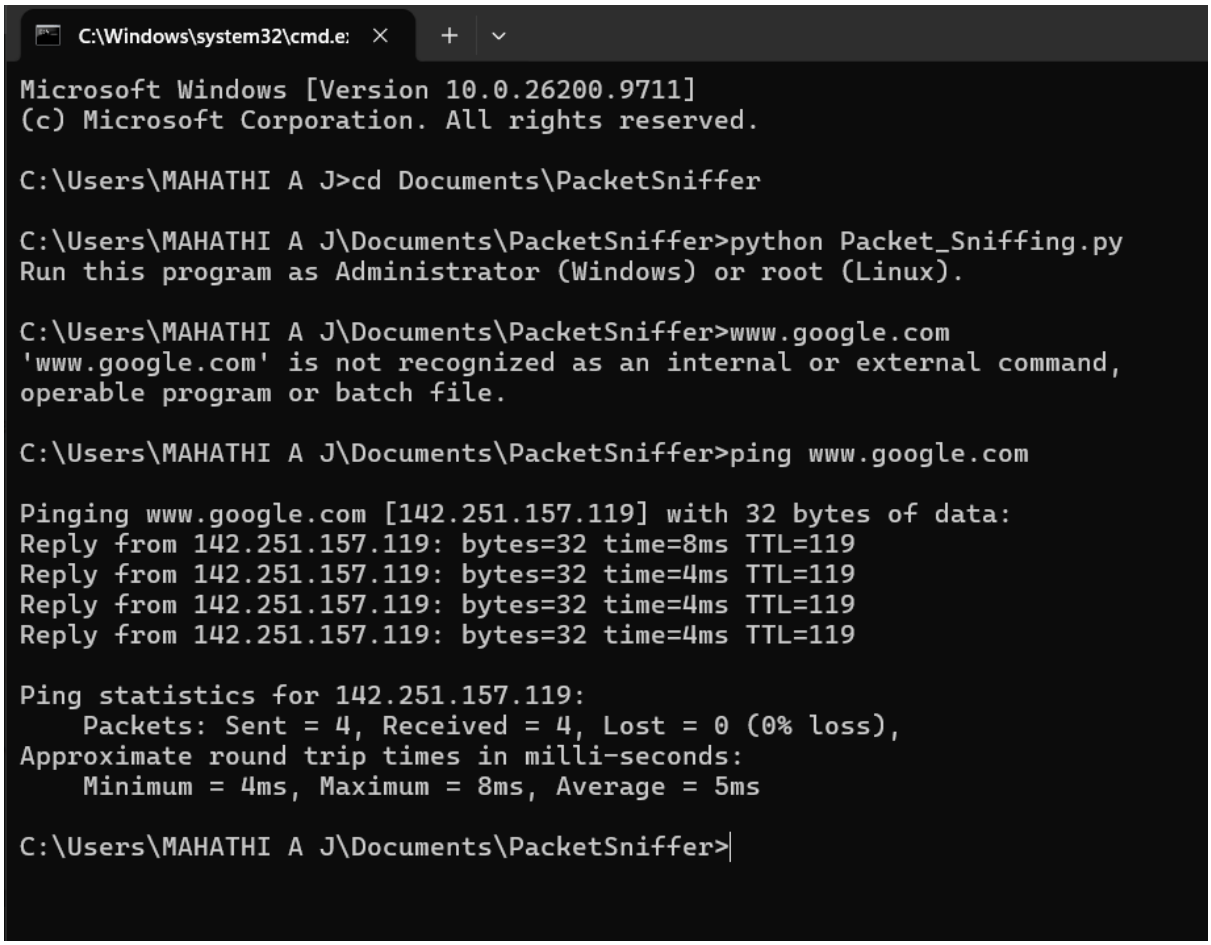
try:
    while True:
        packet, addr = s.recvfrom(65565)

        print("Packet Received")
        print("Source IP:", addr[0])
        print("Packet Size:", len(packet), "bytes")
        print("-" * 50)

except KeyboardInterrupt:
    print("\nStopping Packet Sniffer...")

finally:
    try:
        s.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    except (AttributeError, OSError):
        pass

    s.close()
    print("Packet Sniffer Stopped.")
```


C:\Windows\system32\cmd.e:
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.
C:\Users\MAHATHI A J>cd Documents\PacketSniffer
C:\Users\MAHATHI A J\Documents\PacketSniffer>python Packet_Sniffing.py
Run this program as Administrator (Windows) or root (Linux).
C:\Users\MAHATHI A J\Documents\PacketSniffer>www.google.com
'www.google.com' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\MAHATHI A J\Documents\PacketSniffer>ping www.google.com
Pinging www.google.com [142.251.157.119] with 32 bytes of data:
Reply from 142.251.157.119: bytes=32 time=8ms TTL=119
Reply from 142.251.157.119: bytes=32 time=4ms TTL=119
Reply from 142.251.157.119: bytes=32 time=4ms TTL=119
Reply from 142.251.157.119: bytes=32 time=4ms TTL=119
Ping statistics for 142.251.157.119:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 8ms, Average = 5ms
C:\Users\MAHATHI A J\Documents\PacketSniffer>